

Documentation Plan for DC-Simulator Project

This document summarizes the written documentation produced for the DC-simulator project. It identifies key files, their purposes, and their current status.

1. README / README2

Purpose

Developer documentation and project introduction.

Content

- Project structure
- Build and run instructions
- Data model definitions (`TrainTimetable` , `PowerPoint` , etc.)
- Example configurations
- Implementation notes and utilities

Status

- Multiple versions
 - `README` : general documentation
 - `README2` : focused on "Basic bus models"
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2. softwareSpecification

Purpose

Functional and technical specification.

Content

- Vision and objectives
- Architecture and component responsibilities
- Use of Akka for event-driven simulation

- Requirements for AC/DC simulation
 - Dynamic topology handling
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3. Presentation Outline (PowerPoint)

Purpose

Presentation material for stakeholders.

Content

1. Project scope and goals
2. Electrical network model
3. Simulation structure
4. Braking and regeneration behavior
5. Simulation results and analysis
6. Plots from `MinimalTest`
7. Power balance (line, train, braking resistor)
8. Achievements and next steps

Status

- Exported as `DC-simulator-presentation.pptx`
 - Extended with simulation plots and Excel/CSV outputs
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4. Patch/version management notes

Purpose

Describe the synchronization and versioning strategy.

Content

- Use of zip-archives for full snapshots (e.g., `v0.2` , `v0.4`)
 - Patch-based updates for consistent sync
 - Git commit structure and rollback points
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Summary

The above documents collectively describe:

- The simulator design and purpose
- How to use and extend the system
- Key results and presentation material
- Version control methodology

Next step: bundle all files into a single archive for distribution or archival.